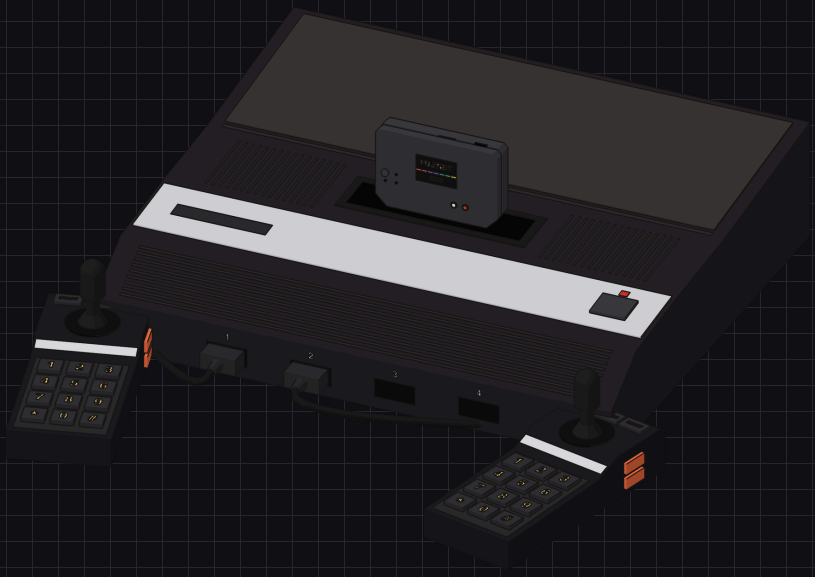
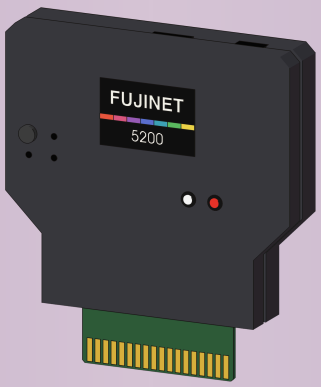
FUJINET 5200

Owner's Manual



FUJINET

A FujiNet Project Publication



FUJINET 5200

FujiNet 5200 Owner's Manual. First edition, 2026.

Typeset after the 1982 *ATARI 5200 Owner's Manual* as a tribute. FujiNet is an independent project and is not affiliated with, endorsed by, or sponsored by Atari. "Atari" and "5200" are trademarks of their owner and are used here to identify the console this manual is about. The games named in it are the FujiNet project's own, except M.U.L.E., a trademark of its owner.

This manual describes the FujiNet cartridge for the ATARI 5200 as it is meant to work. The cartridge is new, and the drawings of it are the manual's own; where today's firmware and games still differ, the differences are listed in the manual's errata, kept with its sources.

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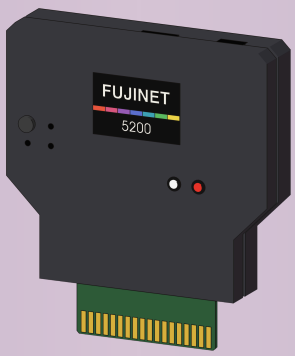
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1. Unpacking Your FujiNet Cartridge

Congratulations on your FujiNet cartridge, one of the most exciting things your ATARI 5200 has ever been plugged into. We suggest that you read this Owner's Manual thoroughly before setting it up.

When you take the FujiNet cartridge from its box, you will have:

- The FujiNet cartridge
- This Owner's Manual



The FujiNet cartridge goes in your console's cartridge slot like any other game. Inside it is a complete network adapter: a WiFi radio, a memory-card reader and two small computers. It lets your 5200 play games that are kept on the Internet, on a memory card, or on a computer in your home – and play some of them against people far away.

What You Also Need

- **An ATARI 5200** of any model: the original four-port console, the later four-port, or the two-port model. FujiNet is made for **NTSC** (North American) consoles and television sets.
- **A 5200 controller**, plugged into **jack 1**. The FujiNet menu is worked with the joystick, the keypad and the side buttons.

- **Your WiFi network's name and password.** FujiNet uses the 2.4 GHz band, which nearly every home router offers.

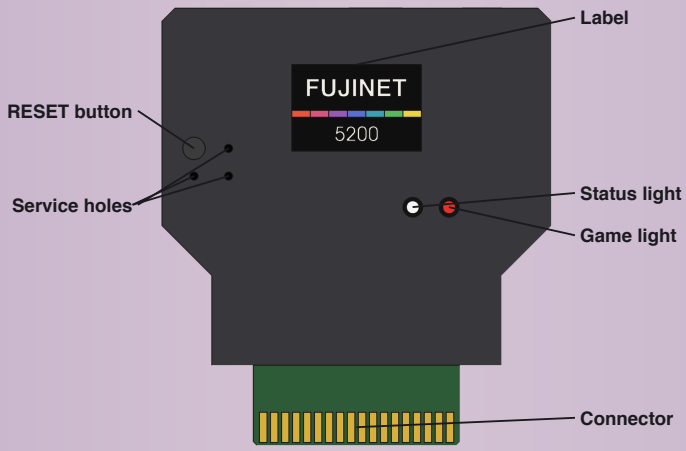
You may also want, though you do not need them to start:

- **A microSD memory card**, formatted FAT32 (most cards come that way), to keep games on.
- **A USB-C phone charger and cable**, for extra power. With a four-port console you will most likely need one; see Section 4.

NOTE: FujiNet makes no changes to your console. Take the cartridge out and your 5200 is exactly as it was.

2. Know Your Cartridge

The cartridge has two faces. The **label** faces you, toward the front of the console, and carries everything you can press or see light up. The other face is plain. The top edge, which stands up out of the console, has the sockets for power and the memory card.



The Lights

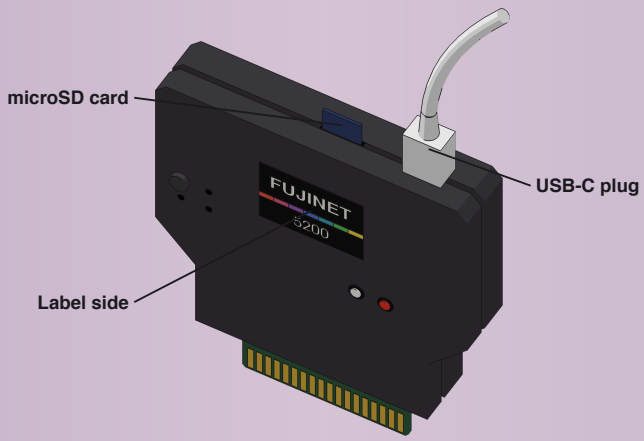
Light	What it means
Status, white	FujiNet is connected to your WiFi network.
Status, orange, flickering	FujiNet is busy: reading the network or the memory card, or sending a game to the console. It goes back to white when it is done.
Status, off	FujiNet is not connected to WiFi (yet).
Game, red	A game has been sent to the console since you turned it on. It is dark while the FujiNet menu is on the screen.

The RESET Button and the Service Holes

The **RESET** button restarts the cartridge itself: both of its small computers. The 5200 cannot be told, so after pressing it, turn the console OFF and ON again. You will rarely need it. Turning the console OFF and ON is the everyday way back to the FujiNet menu.

IMPORTANT: The three small **service holes** beside RESET are for loading new firmware into the cartridge. Do not push anything into them in everyday use.

The Top Edge

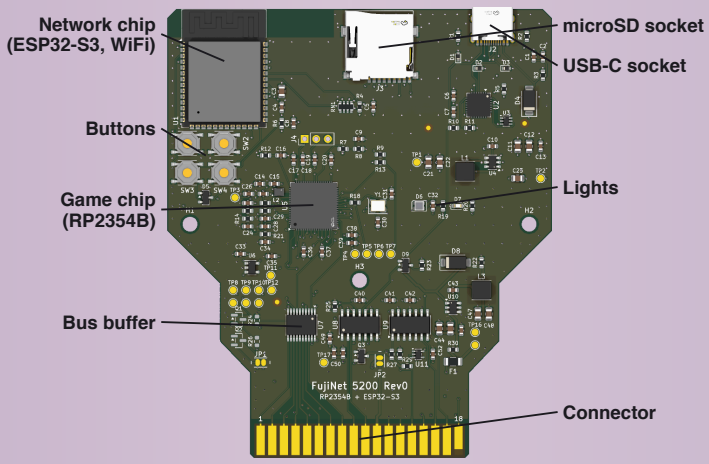


USB-C socket. The cartridge can run on power from the console, and a USB-C phone charger plugged in here powers it too (Section 4). Power from the charger never flows back into the console, so it is safe to leave it plugged in. The same socket is used to load new firmware.

Memory card slot. A microSD card pushes into the slot beside the USB-C socket, contacts first and its printed side toward the label. Push it in until it clicks; push again to let it out.

A Look Inside

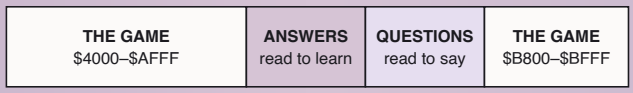
You never need to open the cartridge, but if you are curious, this is the board inside it, seen from the label side.



The FujiNet 5200 board, Rev0. The network chip does the WiFi, the memory card and the talking to servers; the game chip answers the console and plays the part of each game's cartridge.

3. How It Works

Your 5200 was built in 1982 to play cartridges. It has no network port, no keyboard and no disk drive, and its connector for a cartridge has no wire that says “write”: the console can **read** a cartridge, but it has no way to send it anything at all. A FujiNet cartridge puts it on the Internet anyway. Here is the trick.



The 32K the console sees in the cartridge slot, while the FujiNet menu or a FujiNet game is running.

The console sees a cartridge as 32,768 bytes it can look at. FujiNet’s game chip watches **every** look: which address the console reads, at the moment it reads it. So FujiNet sets aside a few pages of that space. Reading in the **questions** pages is how the console speaks – the address it reads spells the byte it means. The **answers** pages are where FujiNet puts what it found out, and the console reads them like any other part of a cartridge.

	The console, 1982	The cartridge, 2026
Computers	one 6502C, about 1.8 million steps a second	two (the game chip and the network chip), each over a hundred times faster
Memory	16,384 bytes	over half a million bytes, and 8 million more beside
Storage	the 32K a cartridge shows it	18 million bytes of its own, and the memory card
Network	none	WiFi, and through it the whole Internet

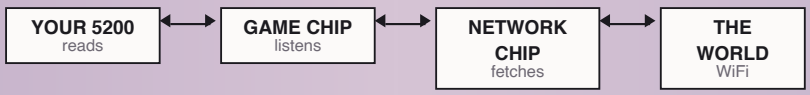
How a Question Gets Answered

1 **ASK**
The program reads its question out of the questions pages, one byte at a time: to say 65, it reads the 65th place in the page. Nothing crosses the connector but the address, and the game chip catches each one as it goes by.

2 **PASS IT ON**
The game chip hands the question to the network chip beside it.

3 **FETCH**
The network chip does the work a 1982 console never could: it joins your WiFi, reaches the server, or reads the memory card.

4 **ANSWER**
The answer goes into the answers pages. To the console it looks as though it had been printed in the cartridge all along. It simply reads it.

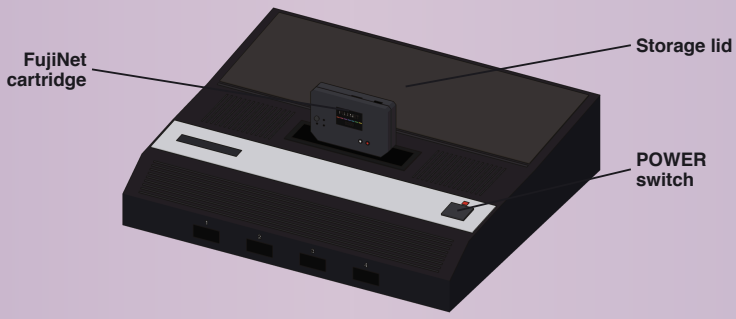


Whole Games

When you load a game, FujiNet sends all of it – up to 131,072 bytes – into the game chip's own memory. Then the menu steps aside and the cartridge becomes that game's cartridge: a plain one, a two-chip one, Bounty Bob Strikes Back's, or a 128K Super Cart, page-switching and all. The console starts the game just as it would at power-on, and never knows the difference.

The full story – every address, every command, and how to write your own FujiNet programs for the 5200 – is in the *FujiNet 5200 Programmer's Handbook*, Sections 3, 4, 13 and 15.

4. Inserting the Cartridge



1. Check to see that the console's **POWER** is **OFF**.
2. Hold the FujiNet cartridge so the label faces you and reads right side up.
3. Carefully insert the cartridge into the cartridge slot in the 5200 console. Be sure the cartridge fits firmly into the slot but DO NOT FORCE IT IN. The top edge, with its USB-C socket, stays standing out of the slot.
4. Plug your controller into **jack 1** (below).
5. Turn your television set on and set the volume control to a comfortable level.
6. Press the console's **POWER ON/OFF** switch. The red light beside it glows, and the FujiNet menu appears.

IMPORTANT: To protect your console and the cartridge, the console's POWER should be OFF whenever you put a cartridge in or take one out.

When you take the cartridge out, pull it straight up, and check first that the POWER is OFF.

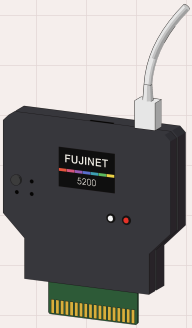
Plugging In the Controller



Plug a 5200 controller into **jack 1** on the front of the console. The FujiNet menu and every FujiNet game listen to the controller in jack 1. The jack is designed to fit only one way. Be sure the connection is firm, but do not try to force it.

NOTE: Some games want a second controller, in jack 2. The game will tell you; FujiNet does not mind what is plugged in.

Extra Power



The cartridge can run on power from the console, but its WiFi radio needs more, in short bursts, than some consoles have to spare. A **USB-C phone charger** plugged into the top of the cartridge supplies it. The cartridge takes what it needs from the charger and never sends any of it back to the console.

- **Four-port consoles.** Use a charger. The original four-port console feeds the cartridge from the same small supply that runs its own memory and controllers, and WiFi can overload it.
- **Two-port consoles** power the cartridge well on their own. Use a charger if the picture jumps or the FujiNet menu starts over by itself when the WiFi connects.

IMPORTANT: Plug the charger into the cartridge **before** you turn the console on. The cartridge is then already awake when the console first looks at it.

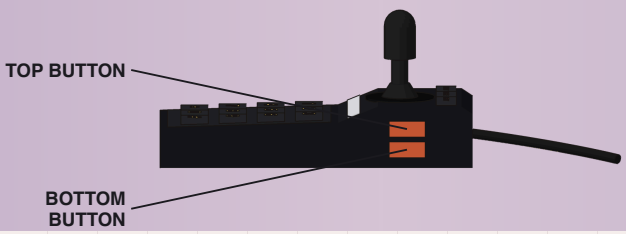
With a charger plugged in, FujiNet stays awake and connected while the console is off, so it is ready again the moment you turn the console back on.

5. Using the 5200 Controller

The FujiNet menu listens to the controller in **jack 1**. It can be worked with the joystick or with the keypad alone, whichever you like better.



Each controller has two buttons on either side. The **bottom buttons** are the fire buttons for all games; in the FujiNet menu they open and choose. The **top buttons** do different things in different games; in the FujiNet menu they go back. The right or the left button can be used, whichever is more comfortable for you.



In the FujiNet Menu

A bar across the screen marks the line you are on. The bottom two lines of every FujiNet screen say which keys do what there, as **#--OPEN** or **PAUSE--RENAME**.

In every FujiNet menu

▲▼◀▶ or 2 4 6 8	Move. Up and down go along the list; left and right turn the page. Hold to repeat.
BOTTOM button or #	Open, or choose, the line the bar is on.
TOP button or *	Go back. On the keyboard, erase a letter.
PAUSE	Rename a host; set a filter; capitals or small letters on the keyboard.
START	Show the adapter's INFO; mark a game to copy; look for networks again; accept what you typed.
RESET	The same as *. It does not restart anything.

The other keypad keys, 1 3 5 7 9 0, are not used by the menu. The games you load decide what every key does.

Getting Back to the FujiNet Menu

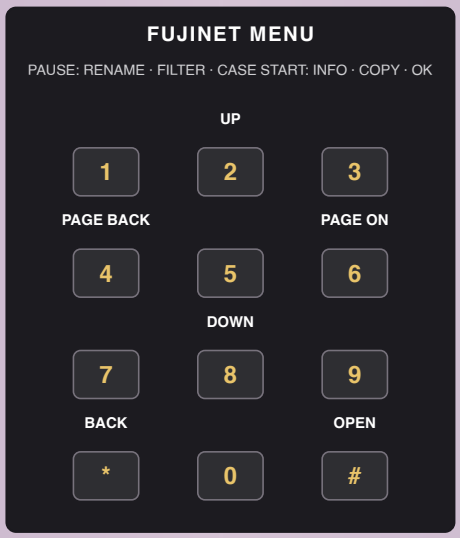
The 5200 has no wire to tell a cartridge to start over, so the way back to the FujiNet menu from any game is:

1. Press the console's **POWER** switch OFF.
2. Wait a moment, and press it ON again.

The FujiNet menu comes back, with your hosts and WiFi just as you left them. The RESET key on the controller belongs to the game you are playing: it starts that game over from the beginning. The FujiNet network games also have a way back of their own; each one's section tells you how.

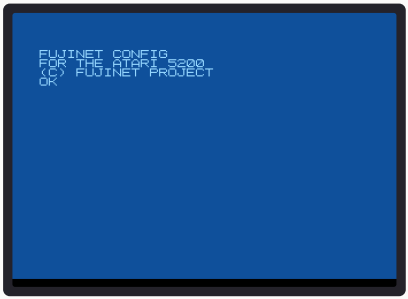
Keypad Overlay

All 5200 games can come with keypad overlays, slipped over the keypad to show what each key does. This is the FujiNet menu's. Copy it onto a blank overlay, or keep this page beside you while you learn.


FUJINET MENU
PAUSE: RENAME · FILTER · CASE START: INFO · COPY · OK
UP
1
2
3
PAGE BACK
PAGE ON
4
5
6
DOWN
7
8
9
BACK
OPEN
*
0
#

6. Join Your Network

Each time you turn the console on, FujiNet types its name across the screen and says **OK**. Press any key to hurry it along.



The FujiNet menu, starting

The first time, FujiNet has no network to join yet, so it shows **SELECT NETWORK** and listens for WiFi networks nearby. **SCANNING...** stays on the screen for a few seconds; then it lists what it heard. After that it remembers your network, joins it by itself, and goes straight to the list of hosts (Section 7).



SELECT NETWORK

1. Move the bar to your network's name.
2. Press the bottom button (or #). The keyboard appears, asking **PASSWORD?**.
3. Spell your network's password (see *The Keyboard*, next).
4. Press START. FujiNet tries to join.

The number at the right of each name is how strongly FujiNet hears that network: the nearer to 0, the stronger. **-42** is strong; **-85** is very faint. Up to fifteen networks are listed, and START looks again. If your network is not in the list – some routers keep their names hidden – choose **OTHER (TYPE AN SSID)** at the bottom. FujiNet asks for the network's name (**SSID?**) first, and then for the password.



Joining



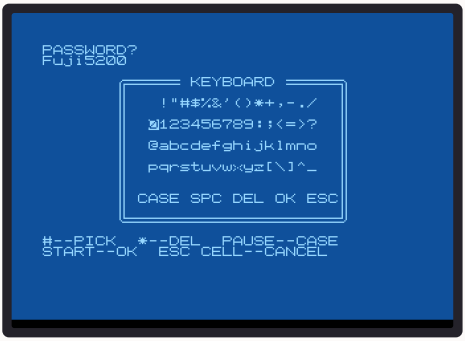
Joined

JOINING NETWORK shows the network's name and **PLEASE WAIT**. It can take up to about forty seconds. When it says **CONNECTED**, the status light on the cartridge turns white and the list of hosts appears. If the password was not right, or the router is too far away, it says why – for example **? CONNECT FAILED** – and goes back to SELECT NETWORK to try again. * while it is joining goes back at once.

NOTE: You can skip WiFi altogether: press * at SELECT NETWORK. FujiNet goes to the list of hosts and still plays games from its memory card.

The Keyboard

FujiNet draws a keyboard whenever it needs words: a password, a network's name, a host's name, or a filter. The top line asks the question; the line under it is what you have typed so far.



The keyboard, with a password typed

On the keyboard

▲▼◀▶ or 2 4 6 8	Move the block from character to character.
BOTTOM button or #	Type the character the block is on, or do what its word says.
TOP button, * or RESET	Erase the last character.
PAUSE	Capital letters or small ones. **CASE** on the bottom row does the same.
START	Accept what you typed. **OK** does the same.

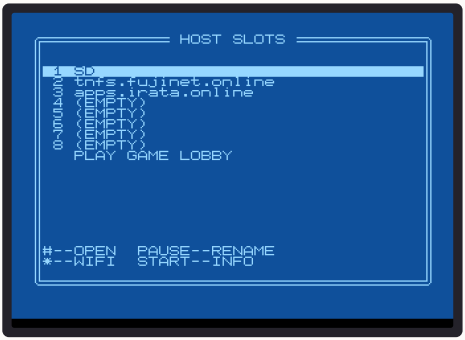
SPC types a space and **DEL** erases. **ESC** is the only way to leave without changing anything. A password can be up to 63 characters, a network's name up to 32.

IMPORTANT: Passwords care about capitals and small letters. Watch the keyboard: after PAUSE its letters are small, and so is everything you type until you press it again.

The keyboard has no **`**, **{**, **|**, **}** or **~**. If your password uses one of them, set your WiFi from a computer instead (Section 10).

7. Hosts and the Browser

A **host** is a place where games are kept: the memory card in the cartridge, a server on the Internet, or a computer in your home. FujiNet remembers eight of them, and lists them on the **HOST SLOTS** screen every time you turn on.



HOST SLOTS, as it comes

A new FujiNet comes with three hosts already filled in:

- **SD** – the microSD card in the top of the cartridge.
- **tnfs.fujinet.online** – the FujiNet project's own library.
- **apps.irata.online** – the library that keeps the 5200's network games.

The other five say **(EMPTY)**. Under them, **PLAY GAME LOBBY** opens the FujiNet Game Lobby (Section 11). Move the bar to a host and press the bottom button (or #) to look inside it.

On HOST SLOTS

BOTTOM or #	Open the host the bar is on (or the Lobby).
PAUSE	Fill in, or change, the host the bar is on.
START	The adapter's INFO (Section 10).
TOP or *	Choose a WiFi network again (Section 6).

Adding a Host

1. Move the bar to an **(EMPTY)** line.
2. Press PAUSE. The keyboard appears, asking **HOST NAME?**.
3. Spell the server's name or address, for example `192.168.1.20` or `games.local`.
4. Press START.

To change a host, rename it the same way. A host name can be up to 31 characters. Most hosts are **TNFS** servers, and a plain name means TNFS. FujiNet can also reach other kinds of server if you spell the kind in front:

Type this	For
`SD`	the memory card in the cartridge
`games.local` or `192.168.1.20`	a TNFS server – the usual kind
`smb://server/share`	a Windows or Samba file share
`nfs://server/export`	an NFS share
`ftp://server`	an FTP server
`http://server/path`	a web server (`https://` too)

The Browser

When you open a host, the browser shows what is in it, sixteen names to a page. The top line says which host you are in (**HOST 3** is the third on HOST SLOTS); the next line is the folder you are in. Folders end in **/** and come first.



The top of *apps.irata.online*



A folder on the memory card

In the browser

▲▼ or 2 8	Move up and down. At the top or bottom of a page, go on to the next one.
◀▶ or 4 6	Turn the page back or on.
BOTTOM or #	On a folder, open it. On a game, **load it** (Section 8).
TOP or *	Back up one folder. At the top of the host, back to HOST SLOTS.
PAUSE	Filter: show only the names that match a pattern.
START	Copy the game the bar is on to another host (Section 9).

Filters. Spell a pattern on the keyboard: `*` stands for any letters and `?` for any one letter, so `B*` shows only the names beginning with B, and `*CARD*` only those with CARD in them. Capitals and small letters match each other, and folders are always shown. An empty filter shows everything again. If nothing matches, the browser says **(NO MATCHES)**.

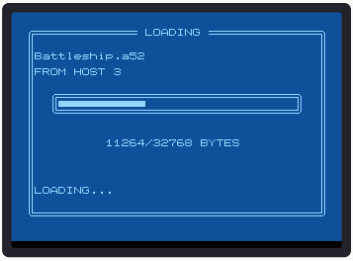
Long names are cut short, with **...** in the middle; the full name is still what FujiNet loads.

8. Loading Games

To load a game, find it in the browser and press the bottom button (or #). FujiNet sends the whole game into the cartridge and starts it – there is nothing else to do.



The network games, on *apps.irata.online*



Loading

LOADING shows the game's name, which host it comes from, and a bar that fills as it loads, with the bytes counted under it. Then it says **BOOTING**, the screen changes, and the game starts just as if its own cartridge were in the slot.

IMPORTANT: To leave a game and go back to the FujiNet menu, turn the console OFF and ON again (Section 5). The RESET key on the controller only starts the game over.

From the Internet

The FujiNet project keeps the 5200's network games on the Internet, ready to load. Nothing needs to be copied:

Host	Folder	What is there
apps.irata.online	**ATARI-5200** ▸ **FUJINET**	The FujiNet network games: Battleship, Five Card Stud, Fujitzee and Texas Hold'em (Sections 12–15)
apps.irata.online	**mule** ▸ **atari5200**	M.U.L.E. (Section 17)
tnfs.fujinet.online		The FujiNet project's library for every computer and console

Open the host, open the folder, and choose the game.

From the Memory Card

1. On your computer, copy your games onto a microSD card formatted FAT32. You may put them in folders.
2. With the console OFF, push the card into the top of the cartridge until it clicks (Section 2).
3. Turn on. Open the **SD** host, and choose a game.

From Your Own File Server

A computer in your home can serve games to FujiNet with a small free program called a **TNFS server**. There are versions for Windows, Mac, Linux and the Raspberry Pi; see **fujinet.online** for the current one.

1. Install the TNFS server, and point it at a folder of games.
2. Find your computer's network address – something like `192.168.1.20`.
3. On HOST SLOTS, rename an **(EMPTY)** host to that address (Section 7).
4. Open it. Your folder appears in the browser.

A Windows share (`smb://`), an NFS share, or an FTP or web server works the same way.

Which Games Will Load

FujiNet loads a game into the cartridge and then plays the part of that game's own cartridge, whatever kind it was. It knows the whole 5200 library by heart, so nearly every game loads just by choosing it:

Kind of cartridge	For example
4K to 32K	Kaboom!, Missile Command, Galaxian, River Raid, H.E.R.O.
16K, made of two chips	Pac-Man, Ms. Pac-Man, Joust, Defender, Qix, Pole Position
Bounty Bob Strikes Back (40K)	the one game of its kind
Super Cart, 64K or 128K	modern homebrews, and M.U.L.E.

File names. FujiNet loads files whose names end in **.A52**, **.BIN** or **.ROM**. A game file ending in `.CAR` loads once it is renamed to end in `.A52`. Games may be up to 128K.

Games FujiNet does not recognise – a new homebrew, say – are judged by their size. If one will not start, a small text file beside it, with the **same name** ending in `.CFG`, can say what kind it is, in one line:

Line	Kind
`mapper=rom`	a plain cartridge, up to 32K
`mapper=2chips`	16K made of two chips
`mapper=bbsb`	Bounty Bob Strikes Back's kind
`mapper=supercart`	a 64K or 128K Super Cart

For example, `NEWGAME.A52` and a file `NEWGAME.CFG` containing just `mapper=2chips`.

Not yet supported: games larger than 128K, and 48K images, which match no kind of cartridge.

NOTE: FujiNet is made for NTSC consoles and television sets. A game made for a PAL machine may roll, or run too fast.

When a Game Will Not Load

Screen says	What it means
?NOT A CARTRIDGE IMAGE	The file is not a 5200 game – or its name does not end in .A52, .BIN or .ROM.
?LOAD ERROR 01	Larger than 128K.
?LOAD ERROR 02	The file is cut short. Copy it again.
?LOAD ERROR 03	A size that matches no kind of cartridge. Try a `.CFG` file beside it.
?MOUNT ERROR, **?PATH ERROR**	FujiNet could not open the host or the file. Check the host is running, and try again.

After a few seconds you are back in the browser.

9. Copying Games

FujiNet can copy a game from one host to another – from the Internet to your memory card, say, so that you can play it without a network.

1. In the browser, move the bar to the game you want to copy and press START.
2. The host list comes back, asking **COPY TO WHICH HOST?**. Move to the host to copy to – **SD** for the memory card – and press the bottom button (or #).
3. Open the folder the game should go in, as in the browser.
4. Press START to copy it there.



Step 2



Step 3

The bottom line says **COPYING...** while it works, and **COPIED** when it is done. Then you are back where you started, in the folder the game came from. To change your mind, press *: on the host list it cancels the copy, and in the folders it backs up one, cancelling at the top.

If it says **?COPY ERROR**, the memory card may be full or missing, or the host may not let you write to it – the Internet libraries, for one, are read-only. Folders cannot be copied, only games, one at a time; a game's `.CFG` file (Section 8), if it has one, is copied the same way, on its own.

10. Info and the Web Control Panel

Press START on HOST SLOTS to see how FujiNet is connected:



ADAPTER INFO

Line	What it is
SSID	the WiFi network FujiNet joined
HOST	FujiNet's own name on your network
IP	FujiNet's address on your network
GATEWAY	your router's address
NETMASK, **DNS**	the rest of your network's settings
MAC	FujiNet's own hardware number
BSSID	the hardware number of the router FujiNet joined
VERSION	the FujiNet firmware version

Press the bottom button, # or * to go back.

The Web Control Panel

Everything FujiNet knows can also be seen and changed from a computer, tablet or phone on the same network. Type FujiNet's **IP** address from the INFO screen into a web browser – for example `http://192.168.1.73/`. On many networks `http://fujinet.local/` works too.

The control panel lets you change the WiFi network, fill in or change the eight hosts with a real keyboard, put games on the memory card, and see the adapter's settings and version. It is the easiest way to type a long host name, or a password the FujiNet keyboard cannot spell.

Setting WiFi from the Memory Card

If you would rather not spell a password with the controller at all, FujiNet can read its settings from the memory card. Make a plain text file called `fnconfig.ini` in the top folder of the card, holding:

```
[WiFi]
enabled=1
SSID=YourNetworkName
passphrase=YourPassword
```

Put the card in the cartridge and turn on. FujiNet reads the file each time it starts. (Everything you change in the menu or the control panel is written back to the same file, so the card keeps your hosts as well.)

11. The Game Lobby

The **FujiNet Game Lobby** is a list of every network game that has a table open right now, on every kind of computer and console FujiNet works with. Pick a table and the Lobby loads the right game for your 5200 and sends you straight to it.

To open the Lobby, move the bar to **PLAY GAME LOBBY** on HOST SLOTS and press the bottom button (or #). FujiNet finds the Lobby's host, **ec.tnfs.io**, and loads the Lobby from it:

- **FINDING LOBBY HOST...** – FujiNet looks for ec.tnfs.io among your hosts.
- **ADDING LOBBY HOST...** – if it is not there, FujiNet puts it in the first **(EMPTY)** host. If all eight are full, it uses the eighth.
- Then **LOADING**, titled **GAME LOBBY**, as for any game (Section 8).

NOTE: The Lobby for the 5200 is being finished now. Until it is ready, load the network games from the browser, as Section 8 shows: each has its own list of tables, and they work just the same without the Lobby. When the games are asked to go back to the Lobby, they go back to the FujiNet menu instead.

Your Name

FujiNet keeps one player name for all of its network games, so you type it once. The first FujiNet game you play asks for it, and every game after that – on this console, and through the Lobby – uses the same one. Names are up to eight letters and numbers. 5 Card Stud, Texas Hold'em and Battleship change it with 4 at their table lists.

Tables

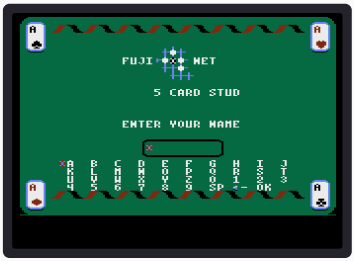
The games' servers keep several tables open all the time. The **AI Rooms** seat you with robot players, which is the best way to learn a game; the others are for people, from wherever they are playing – on a 5200, an Atari computer, an Apple, a PC or anything else FujiNet plugs into.

12. 5 Card Stud

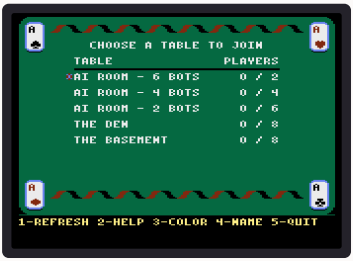
Up to eight players around one table, on any computer or console FujiNet plays on – and if there are not enough people, the house's robots fill the seats. Each hand you get one card face down and four face up, and the best poker hand takes the pot.

Loading the Game

Load it from the browser: host **apps.irata.online**, folder **ATARI-5200 ▸ FUJINET**, game **Five Card Stud.a52**. The first time, the game asks **ENTER YOUR NAME**: move around the letters with the joystick, press the bottom button to type, and choose **OK** when you are done. If you have already given FujiNet a name, it uses that one.



Your name



The tables

Then comes the list of tables. **The Basement** and **The Den** are for people; the **AI Rooms** seat you with two, four or six robot players, which is the best way to learn. Move to a table and press the bottom button to sit down.



Your turn: the bar is on CALL, and 33 seconds are left

The Table

Each seat shows the player's chips over their name, and their cards; your own seat is at the bottom, marked **YOU**. The pot is in the middle, with what each player last did beside their seat – **POST**, **CALL**, **FOLD**. When it is your turn, your moves are on the bottom line, with a bar under the one you are on and the seconds you have left at the right. Otherwise it says whose turn it is.

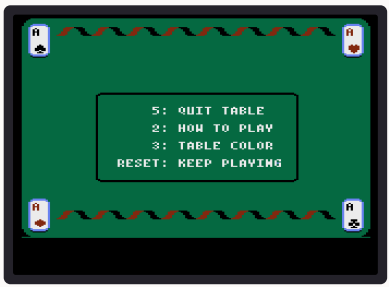
5 Card Stud

▲▼◀▶	Choose a table; at your turn, move the bar along your moves.
BOTTOM, # or START	Sit down; make the move the bar is on.
TOP, * or RESET	The table menu (below).
1	Fetch the table list again.
2 or PAUSE	How to play.
3	Change the colour of the table.
4	Change your name.
5	At the table list: leave the game, for the Lobby (Section 11).

Your First Hand

1. Everyone puts **1** in the pot to start (the ante). Each player is dealt one card face down and one face up.
2. The player showing the lowest card must **POST 2** to begin the betting. Going round the table, each player may **CALL** (match the bet), **BET** or **RAISE** (make it bigger), or **FOLD** (drop out of this hand).
3. A third, fourth and fifth card are dealt face up, each followed by another round of betting. Bets are **5** in the early rounds and **10** later; a round allows no more than three raises.
4. When the betting is done, the hidden cards are turned over. The best poker hand wins the pot.
5. A few seconds later the next hand is dealt.

Everyone starts with **200** chips. You have about **forty seconds** for each move; if the clock runs out, the move under the bar is made for you.



The table menu

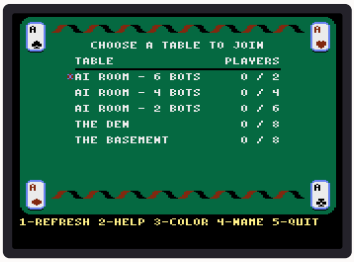
To leave, open the table menu and press 5: **QUIT TABLE** gives your seat back properly, so the others are not left waiting for you, and takes you to the table list. 2 explains the game, 3 changes the table's colour, and RESET goes back to the game.

13. Texas Hold'em

The other great poker game: two cards of your own, five shared by the whole table, and the best five-card hand wins. It plays like 5 Card Stud, on the same kind of table, with robots to fill the seats.

Loading the Game

Load it from the browser: host **apps.irata.online**, folder **ATARI-5200 ▸ FUJINET**, game **Texas Hold'em.a52**. The name screen, the table list and the tables – **The Basement**, **The Den** and the three **AI Rooms** – work as in 5 Card Stud (Section 12).



The tables



After the flop

The three shared cards in the middle are the **flop**, with the round named beside them. Your two cards are at the bottom, by **YOU** your moves – here **FOLD**, **CHECK/BET** and **ALL-IN** – are on the bottom line, with the seconds you have left.

Texas Hold'em

▲▼◀▶	Choose a table; at your turn, move the bar along your moves.
BOTTOM, # or START	Sit down; make the move the bar is on.
TOP, *, PAUSE or RESET	The table menu; PAUSE there goes back to the game.
1 to 5	As in 5 Card Stud: list again, how to play, colour, name, leave.

Your First Hand

1. Two players put in the **blinds**, **5** and **10**, to start the pot. Everyone is dealt two cards face down.
2. Going round the table, each player may **CALL**, **RAISE** or **FOLD** – or **CHECK**, if there is nothing to call.
3. Three shared cards are dealt (the **flop**), then another betting round; then a fourth (the **turn**) and a fifth (the **river**), each with its own betting round.
4. At the **showdown**, each player makes the best five-card hand from their two cards and the five shared ones. The best hand takes the pot.

Everyone starts with **1,000** chips. As in 5 Card Stud you have about forty seconds a move, and the next hand follows by itself. To leave, open the table menu and press 5, **QUIT TABLE**.

14. Battleship

Two to four fleets on one ocean. Hide your five ships, then take turns firing – and every shot you fire lands on **every** enemy board at once. The last fleet afloat wins.

Loading the Game

Load it from the browser: host **apps.irata.online**, folder **ATARI-5200 ▸ FUJINET**, game **Battleship.a52**. Give your name if it asks. The list of games shows each table and how many of its four seats are taken; the bottom line greets you by name. The **AI** tables give you one, two or three robot admirals; **Cape Fuji** and **High Seas** are for people. Move to a table and press the bottom button to join.



The tables



Hiding your ships

In the waiting room, press the bottom button when you are ready (**PRESS FIRE WHEN READY**). The game starts when everyone is ready.

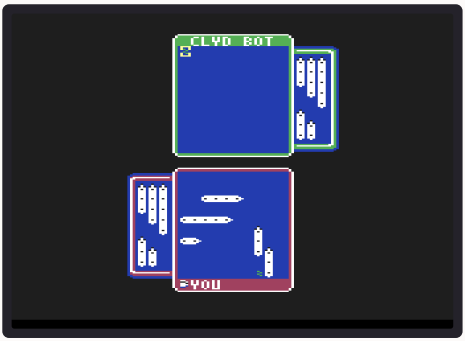
Hiding Your Ships

You have five ships, five, four, three, three and two squares long. The game starts you with a fleet already laid out, so you can simply press the bottom button five times to keep it. To move a ship instead:

Placing ships

▲▼◀▶	Move the ship that is being placed.
TOP or 1	Turn it. (The screen calls the top button **LOWER FIRE**.)
BOTTOM, # or START	Leave it there and go on to the next ship.

Battle



Battle: two fleets, your board at the bottom

Every board is on the screen: yours at the bottom, marked **YOU**, and each enemy's with its admiral's name. Hits and misses are marked on the boards, and each fleet's ships stand beside its board, afloat or sunk. When it is your turn, aim at any square and fire.

In battle

▲▼◀▶	Aim.
BOTTOM, # or START	Fire at that square, on every enemy board at once.
TOP or 1	Ask the server now, instead of waiting.
PAUSE or *	The game menu: 5 quit the game, 2 how to play, 3 sound on or off; the bottom button closes it.
2, 3, 4	How to play, sound, your name.

You have about **45 seconds** a turn. A square you have already fired at on every board is refused, so you cannot waste a shot. The last fleet with a ship afloat wins.

To leave, open the game menu and press 5. At the list of games, 5 leaves Battleship for the Lobby (Section 11).

15. Fujitzee

The dice game. Roll five dice up to three times a turn, keep the ones you like, and score the result in one of thirteen boxes on your card. After thirteen rounds the highest total wins. Up to six players, with robots on hand.

Loading the Game

Load it from the browser: host **apps.irata.online**, folder **ATARI-5200 ▸ FUJINET**, game **Fujitzee.a52**. The very first time, it shows how to play; press any key to read on. Give your name if it asks. On **choose a game to join**, pick a table and press the bottom button. **The Bar** and **Kitchen Table** are for people; the **AI Rooms** add two or four robots.

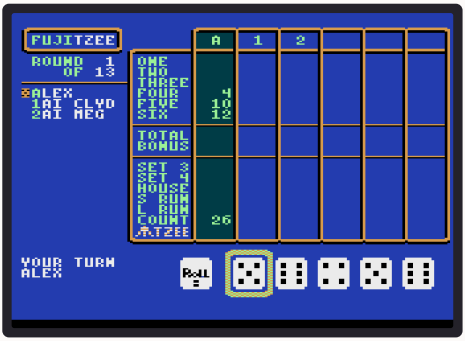


The tables



The waiting room

In the waiting room, press the bottom button to say you are **ready** (**press fire to toggle**). The game starts when everyone is ready.



Your turn, after the first roll

Your card fills the left of the screen, a column for each player, with the round and the players beside it. The five dice sit at the bottom, after **ROLL**.

1. The pointer starts on **ROLL**. Press the bottom button to roll.
2. Move left and right along the dice. The bottom button **keeps** a die; press again to let it go.
3. Roll again – up to three rolls in all.
4. Move up onto the card. Each box shows what this roll would score there. Move to the box you want and press the bottom button.

Fujitzee

▲▼◀▶ or 2 4 6 8	Along the dice; up onto the card; up and down the boxes.
BOTTOM or #	Roll, keep a die, or score a box. In the waiting room, **ready**.
PAUSE or *	The menu: RESET quit the table, 1 how to play, 0 sound; PAUSE keeps playing.
TOP or 1	How to play.
START	At the table list: fetch it again.
RESET	At the table list: leave Fujitzee, for the Lobby (Section 11).

The Card

Box	Scores
one through **six**	The total of the dice showing that number
total, **bonus**	Your top-half total, and a **bonus of 35** once it reaches 63
set 3, **set 4**	Three or four of a kind: the total of all five dice
house	Three of one number and two of another: **25**
s run	Four in a row: **30**
l run	Five in a row: **40**
count	Anything: the total of all five dice
fujitzee	All five the same: **50**

You have about **45 seconds** a turn; if time runs out, the game scores for you (**you timed out. auto scoring.**). After thirteen rounds the standings show who won.

16. Fujirkle

A dice game of nerve. Roll six dice, set aside the ones that score, and either roll on for more or bank what you have – but roll nothing that scores and you lose the whole turn. The first to 10,000 points starts the final round. Up to six players, with robots on hand.

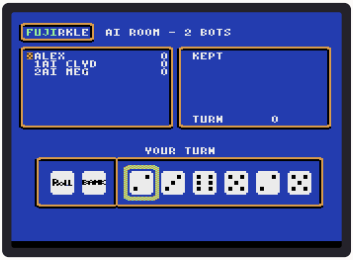
NOTE: Fujirkle for the 5200 is being finished now, and is not yet in the library on *apps.irata.online*. When it is, it will sit beside the other games in **ATARI-5200 ▸ FUJINET**. What follows is how it plays.

Loading the Game

The table list (**choose a game to join**), the name and the waiting room work as in Fujitzee (Section 15): **The Bar** and **Kitchen Table** are for people, the **AI Rooms** add two or four robots, and the bottom button says you are ready.



The tables



Your turn

Your Turn

The players and their scores are at the top left; the dice you have **kept** this turn, and the turn's score so far, at the top right. Your six dice are at the bottom, after **ROLL** and **BANK**.

1. Press the bottom button on **ROLL** to throw the dice.
2. Move along the dice and press the bottom button on each one that scores to set it aside. You must keep at least one.

3. Then either **ROLL** the rest again for more, or **BANK** to add the turn's score to your total and end your turn.
4. Roll nothing that scores, and the turn is over with nothing banked. Set aside all six, and you have **hot dice**: roll all six again.

Fujirkle

▲▼◀▶ or 2 4 6 8	Along the dice. Up jumps to ROLL, down to BANK.
BOTTOM, # or 5	Roll, keep a die, or bank. In the waiting room, **ready**.
START, PAUSE or *	The menu: RESET quit the table, 1 how to play, 3 sound; START goes back to the game.
TOP or 1	How to play.
3	Sound on or off.
0	At the table list: fetch it again.
RESET	At the table list: leave Fujirkle, for the Lobby (Section 11).

What Scores

Dice	Points
A single 1	100
A single 5	50
Three of a kind	100 × the number (three 1s: 1,000)
Four, five or six of a kind	twice, four times or eight times that
1-2-3-4-5-6, a straight	1,500
Three pairs	1,500

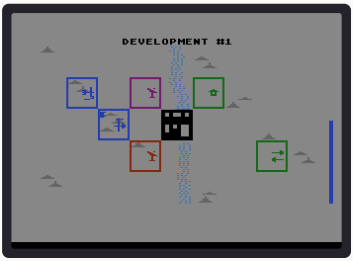
When someone reaches **10,000**, everyone else has one last turn, and the highest score wins. If time runs out on your turn, the game banks for you.

17. M.U.L.E.

The classic game of settling the planet Irata, for four colonists on any FujiNet machine at all – the other three may be on an Atari computer, a 7800, an Apple, a PC or a 2600, and the computer takes any empty seat. Twelve months to claim land, outfit your M.U.L.E.s, and trade your way to the richest colony.



The title



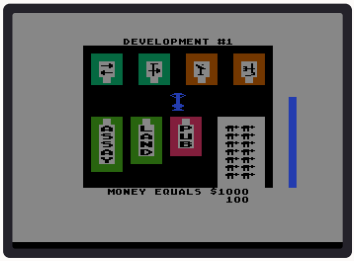
The map

Loading the Game

Load it from the browser: host **apps.irata.online**, folder **mule** ▸ **atari5200**, game **mule.a52** – a 128K Super Cart. On the title screen press the bottom button, give your name if it asks, and pick a table (**UP/DOWN, FIRE**); **Planet IRATA** is the public colony. In the colony room, choose your species with the joystick and press the bottom button when you are ready.



The colony room



The town

M.U.L.E.

▲▼◀▶	Walk; choose; move your auction line; declare.
BOTTOM	Start, join and ready; claim a plot; begin your turn; install a M.U.L.E.; and every other “press the button” of the game.
START	End your turn. On the name screen, done.
TOP	On the name screen, rub out a letter.
0	Sit out a declaration.
#	Sound off and on.
*	Leave the table: press it twice (**LEAVE GAME? * AGAIN: YES**). The computer takes over your colonist. At the table list, * goes back to the FujiNet menu.

The whole game – the colonists, the land, every phase of a month, the auctions, every fortune and catastrophe – is in the *M.U.L.E. Player’s Guide* from the FujiNet project. The 5200 plays like its Atari computer edition.

18. Maintaining Your FujiNet Cartridge

Your FujiNet cartridge should bring many years of enjoyment. To keep it in good working condition, please remember the following points:

- Always turn the console's POWER OFF before putting the cartridge in or taking it out.
- Do not try to force the cartridge into the console.
- Do not push anything into the three small service holes on the cartridge.
- Push the memory card straight in and let it spring straight out. Do not pry at it.
- Use only a USB-C phone charger or a computer's USB port in the USB-C socket.
- Do not spill liquids on the cartridge or pour them into the console's slot.
- Do not drop the cartridge or leave it in great heat.
- Clean the outside of the cartridge with a soft, slightly damp cloth.

Keeping FujiNet Up to Date

FujiNet's firmware – the programs inside the cartridge – is improved all the time. New versions, and how to load them, are announced at **fujinet.online**. Loading new firmware does not change your hosts or your WiFi settings.

19. Troubleshooting Checklist

The following list of symptoms, causes, and remedies is supplied to help you find the trouble when your FujiNet does not do what you expect. As a rule of thumb, if something does not work, turn the console OFF, wait ten seconds, and turn it ON again.

Symptom	• No picture, or a grey or snowy screen.
Probable Cause and Remedy	• Console POWER not ON, or the TV not on the console's channel. • Cartridge not firmly seated. Turn the console OFF, push the cartridge in firmly, and turn it ON. • Cartridge put in while the console was ON. Turn the POWER OFF and back ON.
Symptom	• The picture jumps when WiFi connects, or the FujiNet menu starts over by itself.
Probable Cause and Remedy	• The console cannot give the cartridge enough power. Plug a USB-C phone charger into the top of the cartridge, before turning the console on (Section 4). On a four-port console, always use one.
Symptom	• The controller does nothing.
Probable Cause and Remedy	• Controller plugged into a jack other than jack 1. Move it to jack 1. • Plug not pushed fully home.

Symptom	• The joystick does nothing, but the keypad works.
Probable Cause and Remedy	• Use keypad keys 2, 4, 6 and 8 to move in the FujiNet menu, and tell the FujiNet project (see Getting Help).
Symptom	• My network is not in the SELECT NETWORK list.
Probable Cause and Remedy	• Router too far away. Move the console nearer, or the router. • The network uses only 5 GHz. FujiNet needs 2.4 GHz. • A hidden network. Choose OTHER (TYPE AN SSID) and spell its name.
Symptom	• ?CONNECT FAILED, or ?UNABLE TO CONNECT.
Probable Cause and Remedy	• Wrong password. Remember PAUSE switches between capitals and small letters (Section 6). • Or set your WiFi from a computer, or from the memory card (Section 10).
Symptom	• The status light is not white.
Probable Cause and Remedy	• FujiNet is not on your WiFi. See the two above.

Symptom	• A host shows nothing, or ?HOST ERROR or ? OPEN ERROR.
Probable Cause and Remedy	• Host name misspelled. Rename it (Section 7). • The server is not running, or your Internet connection is down. • SD: no memory card, or it is not formatted FAT32.
Symptom	• ?NOT A CARTRIDGE IMAGE.
Probable Cause and Remedy	• The file's name does not end in .A52, .BIN or .ROM. Rename a .CAR file to .A52 (Section 8). • The file is not a 5200 game.
Symptom	• ?LOAD ERROR 01, 02 or 03.
Probable Cause and Remedy	• A game larger than 128K, a damaged file, or a size FujiNet cannot place. See Section 8.
Symptom	• The game starts, but is garbled or rolls.
Probable Cause and Remedy	• A PAL game. Use the NTSC version. • Try a .CFG file naming the kind of cartridge (Section 8).
Symptom	• I cannot get back to the FujiNet menu.
Probable Cause and Remedy	• Turn the console OFF and ON again. The RESET key on the controller only starts the game over.

Symptom	• A network game says LOBBY NOT AVAILABLE.
Probable Cause and Remedy	• The 5200 Lobby is still being finished (Section 11). Load the games from the browser.
Symptom	• An error with a number, such as ?READ ERROR 01.
Probable Cause and Remedy	• 01: the cartridge's two computers could not talk. Turn the console OFF and ON. • 02: FujiNet took too long to answer. Try again; check the host.

Getting Help

FujiNet is made by a worldwide community of volunteers, and they are glad to help:

- **fujinet.online** – news, downloads and the FujiNet wiki
- **github.com/FujiNetWIFI** – the source for everything, and the place to report a problem
- **discord.gg/7MfFTvD** – the FujiNet Discord, where you can ask anything

20. Parts List

The FujiNet cartridge is open hardware: the board and every program in it may be built, changed and shared. Its design files are in **fujinet-hardware** on GitHub, under **ATARI-5200 / FujiNet-5200-Rev0**. When asking about parts, give the board's name and revision, printed on the board.

Board	FujiNet-5200 Rev0
Game chip	RP2354B (RP2350 with 2 MB flash)
Network chip	ESP32-S3-WROOM-1, 16 MB flash, 8 MB PSRAM
Bus buffer	74HCT541
WiFi	802.11 b/g/n, 2.4 GHz
Memory card	microSD, push-push, FAT32
USB	USB-C: power, and firmware loading
Lights	status (white / orange); game (red)
Buttons	RESET; three service buttons (BOOTSEL, S3 EN, S3 BOOT)
Power	from the cartridge slot, or USB-C
Consoles	CX5200, four-port and two-port; NTSC
Games	up to 128K; .A52, .BIN or .ROM
Cartridge shell	to come; the drawings in this manual are its own

Related Publications

- *FujiNet 5200 Programmer's Handbook* – writing programs for the cartridge.
- *M.U.L.E. Player's Guide* – the whole of M.U.L.E.
- *The FujiNet Network Protocol Handbook* – everything FujiNet can talk to.

A Few Words

Word	Meaning
AI Room	A game table where robot players fill the empty seats.
Firmware	The programs built into the cartridge itself.
FAT32	The way most memory cards are formatted; the one FujiNet reads.
Host	A place games are kept: the memory card, or a server on the Internet or in your home.
Lobby	FujiNet's list of open game tables, on every kind of machine.
NTSC, **PAL**	The North American and the European ways of making a TV picture. FujiNet and its games are NTSC.
SSID	Your WiFi network's name.
Super Cart	A cartridge bigger than the console's 32K window, showing it one part at a time. FujiNet plays the part for the games it loads.
TNFS	The simple file-sharing language FujiNet servers speak.
Two-chip	A 16K cartridge built from two 8K chips. FujiNet knows which games were made this way.
USB-C	The small, either-way-up plug of modern phone chargers.

My FujiNet

Write down your settings here. If you ever start over, everything you need is on this page.

WiFi network (SSID) _____

Password _____

FujiNet's IP address _____

Firmware version _____

Player name _____

Hosts

1 _____

2 _____

3 _____

4 _____

5 _____

6 _____

7 _____

8 _____

Games I like, and where they are

_____

_____

Quick Reference

Screen	Stick or 2 4 6 8	Bottom or #	Top or *	PAUSE · START
The FujiNet menu				
SELECT NETWORK	choose	this one	skip WiFi	– · look again
Keyboard	move	type	erase	case · OK
HOST SLOTS	choose	open	WiFi	rename · INFO
Browser	choose; left/right: page	open · load	up a folder	filter · copy
COPY TO …	choose	pick · open	cancel · back	– · copy here
The games				
5 Card Stud, Hold'em	choose; your move	sit · move	menu	help or menu · move
Battleship	aim; move a ship	place · fire	turn a ship	menu · fire
Fujitzee	dice; the card	roll · keep · score	help	menu · list again
Fujirkle	dice; up ROLL, down BANK	roll · keep · bank	help	menu · menu
M.U.L.E.	walk · choose · bid	the button	rub out	– · end turn
Everywhere				
Console POWER	OFF and ON: back to the FujiNet menu, from any game.			
RESET key	In the menu, the same as *; in a game, the game's own.			

Help: **fujinet.online** · **github.com/FujiNetWIFI** · **discord.gg/7MfFTvD**

FUJINET

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The FujiNet Project
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